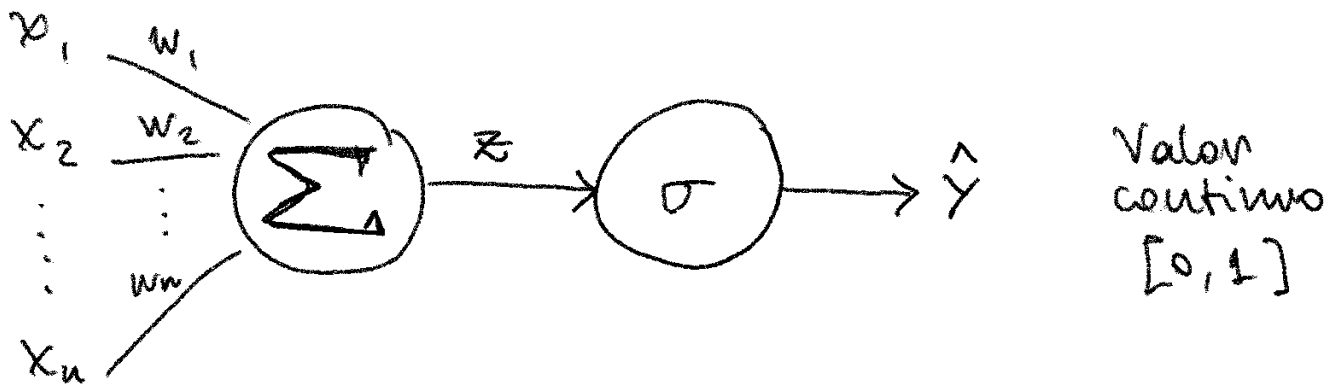


REGRESIÓN LOGÍSTICA



$$x_1 w_1 + x_2 w_2 + \dots + x_n w_n = z$$

$$[w_1, w_2, \dots, w_n] \cdot \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = z = W^T x$$

$$z = Wx$$

$$\hat{y} = \sigma(z)$$

SIGMOIDEA

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

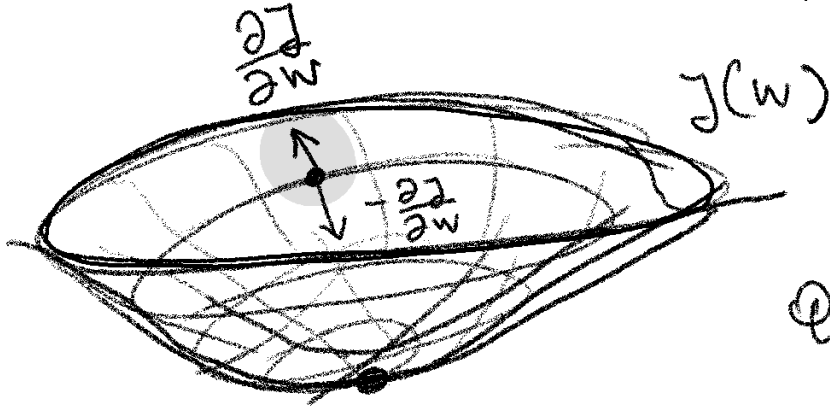
La función de costo :

$$J(w) = -\frac{1}{n} \sum_{i=1}^n \mathcal{L}(y_i, \hat{y}_i)$$

FUNCIÓN DE
COSTO

$$\mathcal{L}(y, \hat{y}) = y \log(\hat{y}) + (1-y) \log(1-\hat{y}) \quad \begin{array}{l} \text{LOSS} \\ \text{PUNTUAL} \\ \text{BCE} \end{array}$$

Yo quiero hallar W tq $J(W)$ sea mínimo.

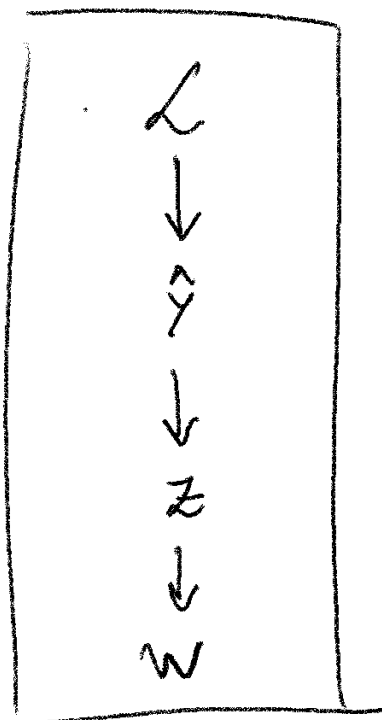


Quiero calcular :

$$\frac{\partial J}{\partial W}$$

$$\frac{\partial J}{\partial W} = -\frac{1}{n} \sum \frac{\partial \mathcal{L}}{\partial W} \overset{\text{BCE}}{\mathcal{L}}(y_i, \hat{y}_i) \quad \text{Necesito } \frac{\partial \mathcal{L}}{\partial W}$$

Como $\mathcal{L}(y, \hat{y})$ y $\hat{y} = \sigma(z)$ $z = Wx$



REGLA
DE LA
CADENA

$$\boxed{\frac{\partial \mathcal{L}}{\partial w} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z} \cdot \frac{\partial z}{\partial w}}$$

$$i) \frac{\partial \mathcal{L}}{\partial \hat{y}} = \frac{\partial}{\partial \hat{y}} \left(y \log(\hat{y}) + (1-y) \log(1-\hat{y}) \right)$$

$$= y \cdot \frac{1}{\hat{y}} + (1-y) \cdot \frac{1}{1-\hat{y}} \cdot (-1)$$

$$= \frac{y}{\hat{y}} - \frac{1-y}{1-\hat{y}} = \frac{y(1-\hat{y}) - \hat{y}(1-y)}{\hat{y}(1-\hat{y})}$$

$$= \frac{y - \cancel{y\hat{y}} - \hat{y} + \cancel{y\hat{y}}}{\hat{y}(1-\hat{y})} = \boxed{\frac{y - \hat{y}}{\hat{y}(1-\hat{y})} = \frac{\partial \mathcal{L}}{\partial \hat{y}}}$$

$$ii) \boxed{\frac{\partial \hat{y}}{\partial z} = \hat{y}(1-\hat{y})}$$

$$\hat{y} = \sigma(z)$$

$$\frac{\partial \hat{y}}{\partial z} = \frac{\partial \sigma}{\partial z} = \sigma(z)(1-\sigma(z))$$

$$iii) \quad z = wx$$

$$\boxed{\frac{\partial z}{\partial w} = x}$$

$$\frac{\partial \mathcal{L}}{\partial w} = \frac{y - \hat{y}}{\hat{y}(1 - \hat{y})} \cdot \hat{y}(1 - \hat{y}) \cdot x = \boxed{(y - \hat{y})x = \frac{\partial \mathcal{L}}{\partial w}}$$

Volvemos a $J(w)$:

$$\boxed{\frac{\partial J}{\partial w} = -\frac{1}{n} \sum (y - \hat{y})x}$$

Nos parecen
exquisitos

$$W = \begin{bmatrix} w_1 \\ \vdots \\ w_j \\ \vdots \\ w_n \end{bmatrix}$$

$$\frac{\partial J}{\partial w_j} = -\frac{1}{m} \sum_{i=1}^m (y^{(i)} - \hat{y}^{(i)}) x_j^{(i)}$$

↓
Pesos de los
features: $j = 1 \dots n$

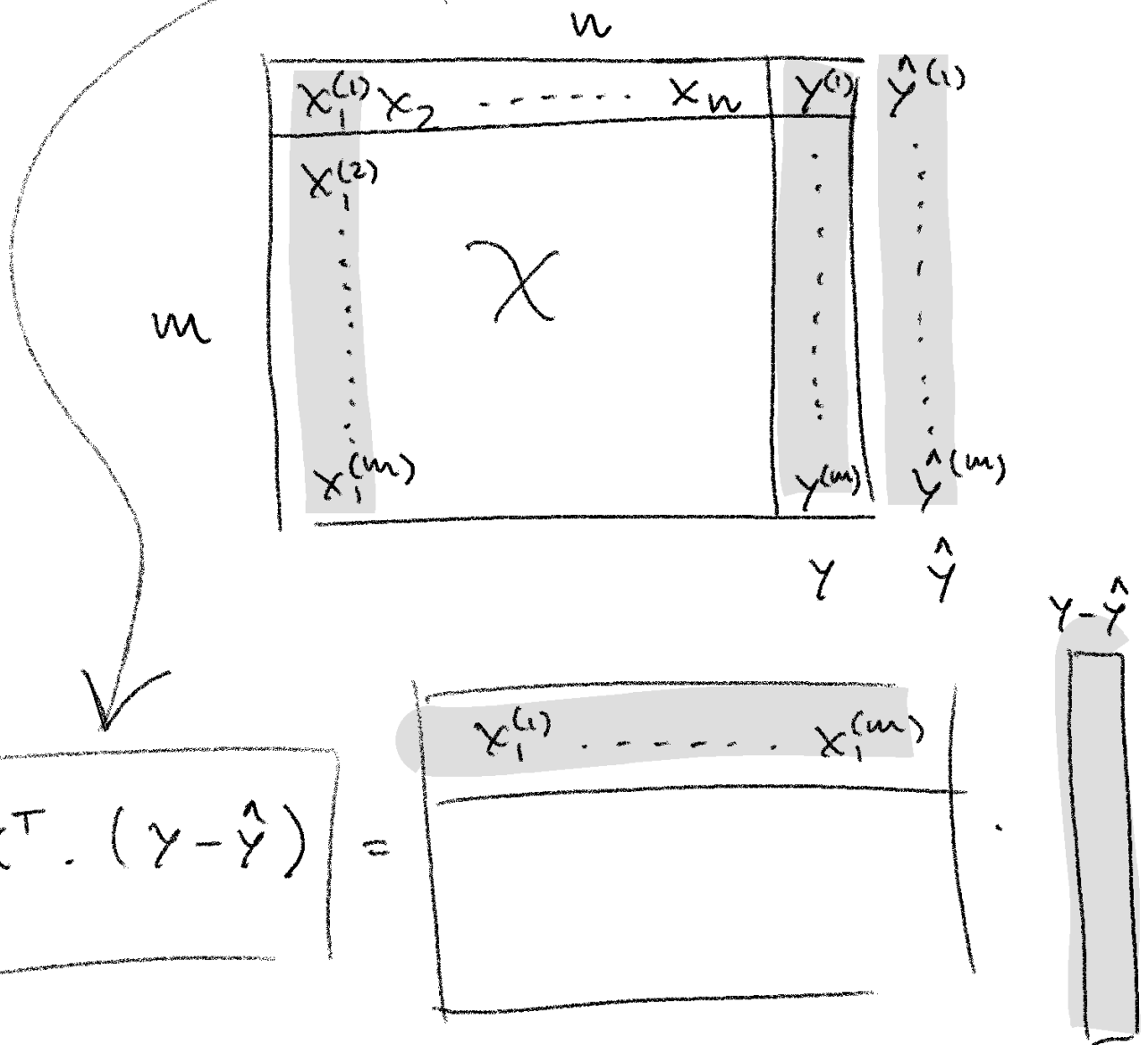
↗ muestras
 $i = 1 \dots m$

Dataset:

	$j = 1 \dots n$					y
	x_1	x_2	...	...	x_n	
$\vdots$	X					
$\vdots$						
m						

FORMA MATRICIAL

$$\frac{\partial J}{\partial w_j} = -\frac{1}{m} \sum_{i=1}^n (y^{(i)} - \hat{y}^{(i)}) x_j^{(i)}$$



Agrego una feature $x_{n+1} = 1$

para modelar el bias : $z = Wx + b$